MANAV RACHNA INTERNATIONAL INSTITUTE OF RESEARCH & STUDIES

SCHOOL OF COMPUTER APPLICATIONS

PRACTICAL FILE

Data Structure and Algorithm

Academic Session: 2026–2027

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Course: MCA

Semester: I

Queue Operations

Code

```
queue = [0] * 5
front = 0
rear = -1
size = 5

while True:
    print("\n===== QUEUE MENU =====")
    print("1. Insert")
    print("2. Delete")
    print("3. Display")
    print("4. Exit")

    choice = int(input("Enter your choice: "))

    match choice:
        case 1:
            if rear == size - 1:
                print("Queue is full")
            else:
                value = int(input("Enter value: "))
                rear += 1
                queue[rear] = value
                print("Element inserted")

        case 2:
            if front > rear:
                print("Queue is empty")
            else:
                print("Deleted element:", queue[front])
                front += 1

        case 3:
            if front > rear:
                print("Queue is empty")
            else:
                print("Queue elements:", end=" ")
                for i in range(front, rear + 1):
                    print(queue[i], end=" ")
                print()

        case 4:
            print("Program closed")
            break

        case _:
            print("Invalid choice")
```

Output Screenshot

Program Output

```
===== QUEUE MENU =====
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value: 10
Element inserted

===== QUEUE MENU =====
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter value: 20
Element inserted

===== QUEUE MENU =====
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 3
Queue elements: 10 20

===== QUEUE MENU =====
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 4
Program closed
```

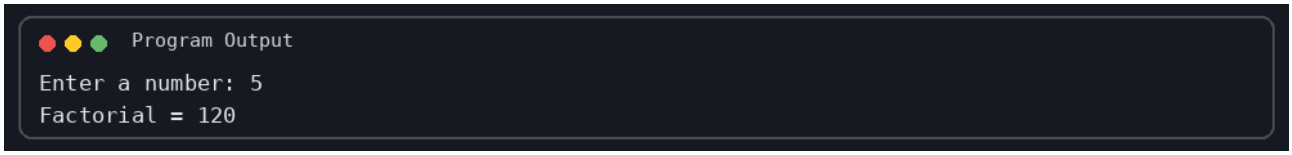
Factorial

Code

```
n = int(input("Enter a number: "))

if n < 0:
    print("Factorial is not defined for negative numbers")
else:
    fact = 1
    for i in range(1, n + 1):
        fact = fact * i
    print("Factorial =", fact)
```

Output Screenshot



Multiplication Table

Code

```
n = int(input("Enter a number: "))

for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

Output Screenshot



Stack Operations

Code

```
stack = [0] * 5
top = -1
size = 5

while True:
    print("\n===== STACK MENU =====")
    print("1. Push")
    print("2. Pop")
    print("3. Peek")
    print("4. Display")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    match choice:
        case 1:
            if top == size - 1:
                print("Stack is full")
            else:
                value = int(input("Enter value: "))
                top += 1
                stack[top] = value
                print("Element pushed")

        case 2:
            if top == -1:
                print("Stack is empty")
            else:
                print("Popped element:", stack[top])
                top -= 1

        case 3:
            if top == -1:
                print("Stack is empty")
            else:
                print("Top element:", stack[top])

        case 4:
            if top == -1:
                print("Stack is empty")
            else:
                print("Stack elements:", end=" ")
                for i in range(top, -1, -1):
                    print(stack[i], end=" ")
                print()

        case 5:
            print("Program closed")
            break

        case _:
            print("Invalid choice")
```

Output Screenshot

Program Output

```
===== STACK MENU =====
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter value: 10
Element pushed

===== STACK MENU =====
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 1
Enter value: 20
Element pushed

===== STACK MENU =====
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 4
Stack elements: 20 10

===== STACK MENU =====
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice: 5
Program closed
```

Deque Operations

Code

```
dq = [0] * 5
start = -1
end = -1
size = 5

while True:
    print("\n===== DEQUE MENU =====")
    print("1. Insert from front")
    print("2. Insert from rear")
    print("3. Delete from front")
    print("4. Delete from rear")
    print("5. Display deque")
    print("6. Exit")

    ch = int(input("Enter your choice: "))

    match ch:
        case 1:
            if end == size - 1:
                print("Deque is full")
            else:
                item = int(input("Enter element: "))

                if start == -1:
                    start = end = 0
                else:
                    k = end
                    while k >= start:
                        dq[k + 1] = dq[k]
                        k -= 1
                    end += 1

                dq[start] = item
                print("Inserted at front")

        case 2:
            if end >= size - 1:
                print("Deque is full")
            else:
                item = int(input("Enter element: "))

                if start == -1:
                    start = 0
                    end = 0
                else:
                    end += 1

                dq[end] = item
                print("Inserted at rear")

        case 3:
            if start == -1:
                print("Deque is empty")
            else:
                print("Deleted item:", dq[start])

                if start == end:
                    start = end = -1
                else:
                    start = start + 1

        case 4:
            if start == -1:
                print("Deque is empty")
            else:
                print("Deleted item:", dq[end])

                if start == end:
                    start = end = -1
                else:
                    end = end - 1

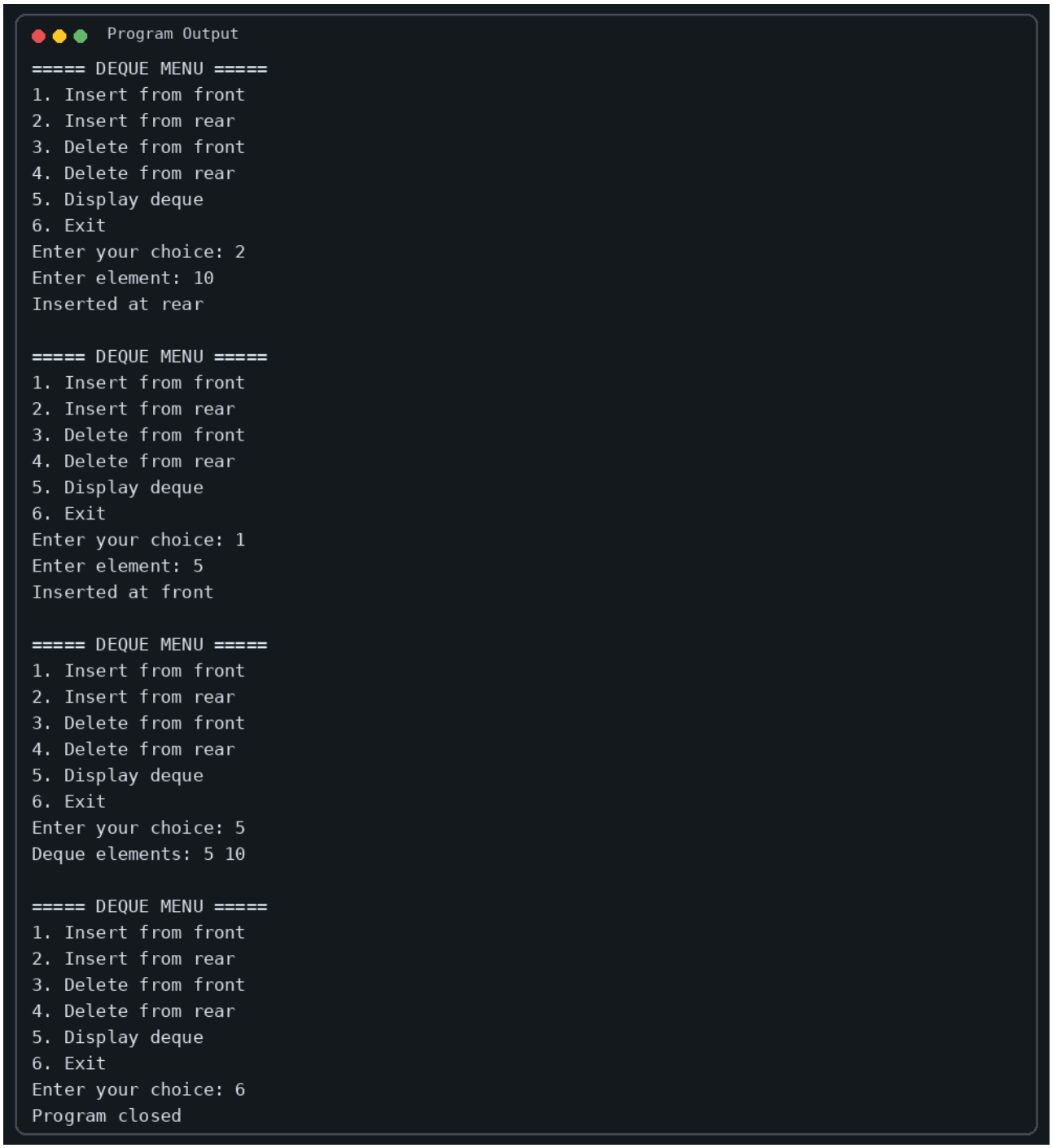
        case 5:
```

```
        if start == -1:
            print("Nothing to display")
        else:
            print("Deque elements:", end=" ")
            for k in range(start, end + 1):
                print(dq[k], end=" ")
            print()

    case 6:
        print("Program closed")
        break

    case _:
        print("Invalid choice")
```

Output Screenshot



Array Operations

Code

```
arr = [0] * 100
n = int(input("Enter number of elements: "))

for i in range(n):
    arr[i] = int(input("Enter value: "))

while True:
    print("\n===== ARRAY MENU =====")
    print("1. Insert element")
    print("2. Delete element")
    print("3. Show array")
    print("4. Exit")

    ch = int(input("Enter your choice: "))

    match ch:
        case 1:
            pos = int(input("Enter position: "))
            num = int(input("Enter element to insert: "))

            if pos > 0 and pos <= n + 1 and n < 100:
                j = n
                while j >= pos:
                    arr[j] = arr[j - 1]
                    j -= 1

                arr[pos - 1] = num
                n += 1
                print("Insertion successful")
            else:
                print("Invalid position or array is full")

        case 2:
            pos = int(input("Enter position to delete: "))

            if pos >= 1 and pos <= n:
                j = pos - 1
                while j < n - 1:
                    arr[j] = arr[j + 1]
                    j += 1

                n -= 1
                print("Deletion successful")
            else:
                print("Invalid position")

        case 3:
            if n == 0:
                print("Array is empty")
            else:
                print("Array elements:", end=" ")
                for i in range(n):
                    print(arr[i], end=" ")
                print()

        case 4:
            print("Program terminated")
            break

        case _:
            print("Wrong choice, try again")
```

Output Screenshot

